Sensor fusion, A2

1a) $E[z] = \Sigma^{-\frac{1}{2}}(E[x] - \mu) = 0$

$Cov(z) = \Sigma^{-\frac{1}{2}}\Sigma(\Sigma^{-\frac{1}{2}})^T = I$

Therefore $z \sim N(0, I)$

b) $z_i \sim N(0,1)$

Let $y_i = z_i^2$

Transformation of random variables for $y_i > 0$

$$p_{y_i}(y_i) = \frac{p_{z_i}(\sqrt{y_i}) + p_{z_i}(-\sqrt{y_i})}{2\sqrt{y_i}} = \frac{1}{\sqrt{2\pi y_i}} e^{-\frac{y_i}{2}}$$

Therefore

$y_i = z_i^2 \sim \chi_1^2$

c) $y_i \sim \chi_1^2$ are independent

$$y = \sum_{i=1}^{n} y_i = \sum_{i=1}^{n} z_i^2$$

MFG of y_i:

$M_{y_i}(t) = (1-2t)^{-\frac{1}{2}}$

$$M_y(t) = \prod_{i=1}^{n} M_{y_i}(t) = (1-2t)^{-\frac{n}{2}}$$

This is MFG of χ_n^2 distribution

Therefore

$y = (x-\mu)^T \Sigma^{-1}(x-\mu) \sim \chi_n^2$

2a) $z^c = H^c x + v^c, \quad v^c \sim N(0, R^c)$

$p(z^c|x) = N(z^c; H^c x, R^c)$

2b) $p(x, z^c) = p(z^c|x)\, p(x)$

Since
$x \sim N(\bar{x}, P), \quad z^c = H^c x + v^c, \quad v^c \sim N(0, R^c)$
and x and v^c are independent

$$\begin{bmatrix} x \\ z^c \end{bmatrix} = \begin{bmatrix} I \\ H^c \end{bmatrix} x + \begin{bmatrix} 0 \\ v^c \end{bmatrix}$$

$$\Rightarrow p(x, z^c) = N\left(\begin{bmatrix} x \\ z^c \end{bmatrix}; \begin{bmatrix} \bar{x} \\ H^c \bar{x} \end{bmatrix}, \begin{bmatrix} P & P(H^c)^T \\ H^c P & H^c P (H^c)^T + R^c \end{bmatrix}\right)$$

2c) $S^c = H^c P (H^c)^T + R^c$

$p(z^c) = N(z^c; H^c \bar{x}, S^c)$

$p(x|z^c) = N(x; \hat{x}^c, P^c)$ where

$\hat{x}^c = \bar{x} + P(H^c)^T (S^c)^{-1} (z^c - H^c \bar{x})$

$P^c = P - P(H^c)^T (S^c)^{-1} H^c P$

2d) $S^r = H^r P (H^r)^T + R^r$

$p(z^r) = N(z^r; H^r \bar{x}, S^r)$

$p(x|z^r) = N(x; \hat{x}^r, P^r)$
where
$\hat{x}^r = \bar{x} + P(H^r)^T (S^r)^{-1} (z^r - H^r \bar{x})$

$P^r = P - P(H^r)^T (S^r)^{-1} H^r P$

2e) $p(x \mid z^c) = N(x; \hat{x}^c, P^c)$

$\hat{x}_{MMSE} = \hat{x}_{MAP} = \hat{x}^c$

$\hat{x}^c = \bar{x} + P(H^c)^T (S^c)^{-1} (z^c - H^c \bar{x})$

3a) $\log [N^{-1}(x; a, B) N^{-1}(y; Cx, D)]$

$= a^T x - \frac{1}{2} x^T B x + (Cx)^T D^{-1} y - \frac{1}{2} y^T D y - \frac{1}{2} x^T C^T D^{-1} C x$

$= \begin{bmatrix} a \\ 0 \end{bmatrix}^T \begin{bmatrix} x \\ y \end{bmatrix} - \frac{1}{2} \begin{bmatrix} x \\ y \end{bmatrix}^T \begin{bmatrix} B + C^T D^{-1} C & -C^T \\ -C & D \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

The product is the canonical Gaussian

$N^{-1}\left(\begin{bmatrix} x \\ y \end{bmatrix}; \begin{bmatrix} a \\ 0 \end{bmatrix}, \begin{bmatrix} B + C^T D^{-1} C & -C^T \\ -C & D \end{bmatrix} \right)$

3b) $\eta_a = a, \quad \eta_b = 0$

and $\Lambda_{xx} = B + C^T D^{-1} C, \quad \Lambda_{xy} = -C^T, \quad \Lambda_{yy} = D$

Therefore

$\eta^* = 0 - (-C)(B + C^T D^{-1} C)^{-1} a = C(B + C^T D^{-1} C)^{-1} a$

and $\Lambda^* = D - (-C)(B + C^T D^{-1} C)^{-1} (-C^T)$

$= D - C(B + C^T D^{-1} C)^{-1} C^T$

Therefore

$p(y) = N^{-1}(y; C(B + C^T D^{-1} C)^{-1} a, D - C(B + C^T D^{-1} C)^{-1} C^T$

3c) $\eta_{x|y} = \eta_a - \Lambda_{xy} y$, $\Lambda_{x|y} = \Lambda_{xx}$

$\eta_a = a$, $\Lambda_{xy} = -C^T$, $\Lambda_{xx} = B + C^T D^{-1} C$

$\eta_{x|y} = a - (-C^T) y = a + C^T y$ and $\Lambda_{x|y} = B + C^T D^{-1} C$

$p(x|y) = N^{-1}(x; a + C^T y, B + C^T D^{-1} C)$

3d) $B = \bar{P}^{-1}$, $C = H$, $D^{-1} = R^{-1}$

$\hat{P}^{-1} = B + C^T D^{-1} C$

Therefore

$\hat{P}^{-1} = \bar{P}^{-1} + H^T R^{-1} H$